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Electronic Structure and Hybridization Effects in Silicon-Doped Graphene systems

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Abstract

Density Functional Theory was employed to analyze hybridization effects in graphene's electronic structure at varying silicon concentrations utilizing Virtual Crystal Approximation (VCA). The results show a correlation between silicon concentrations and electronic structure near the Fermi-level, specifically a silicon-induced modification of the density of states.

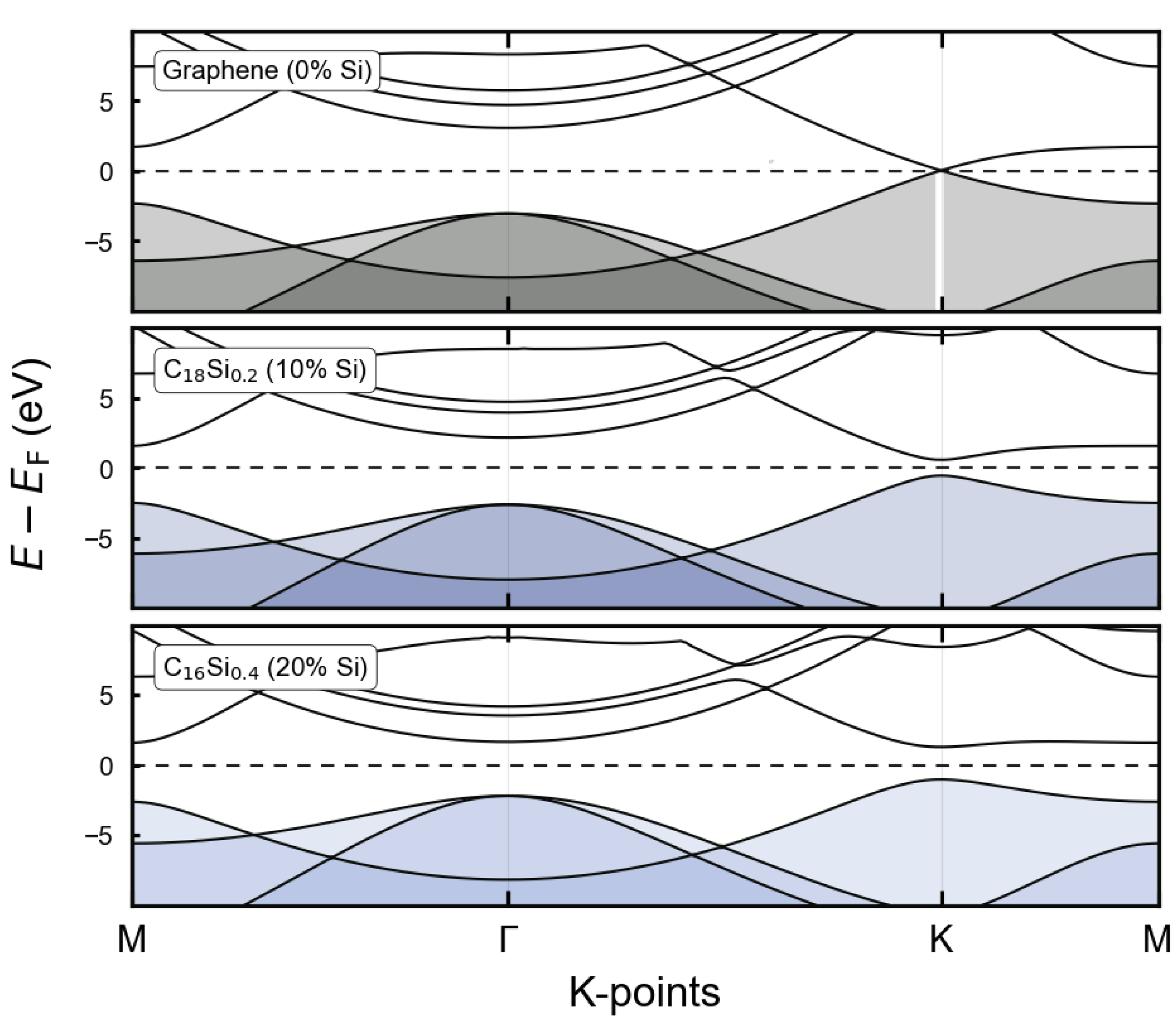
Introduction

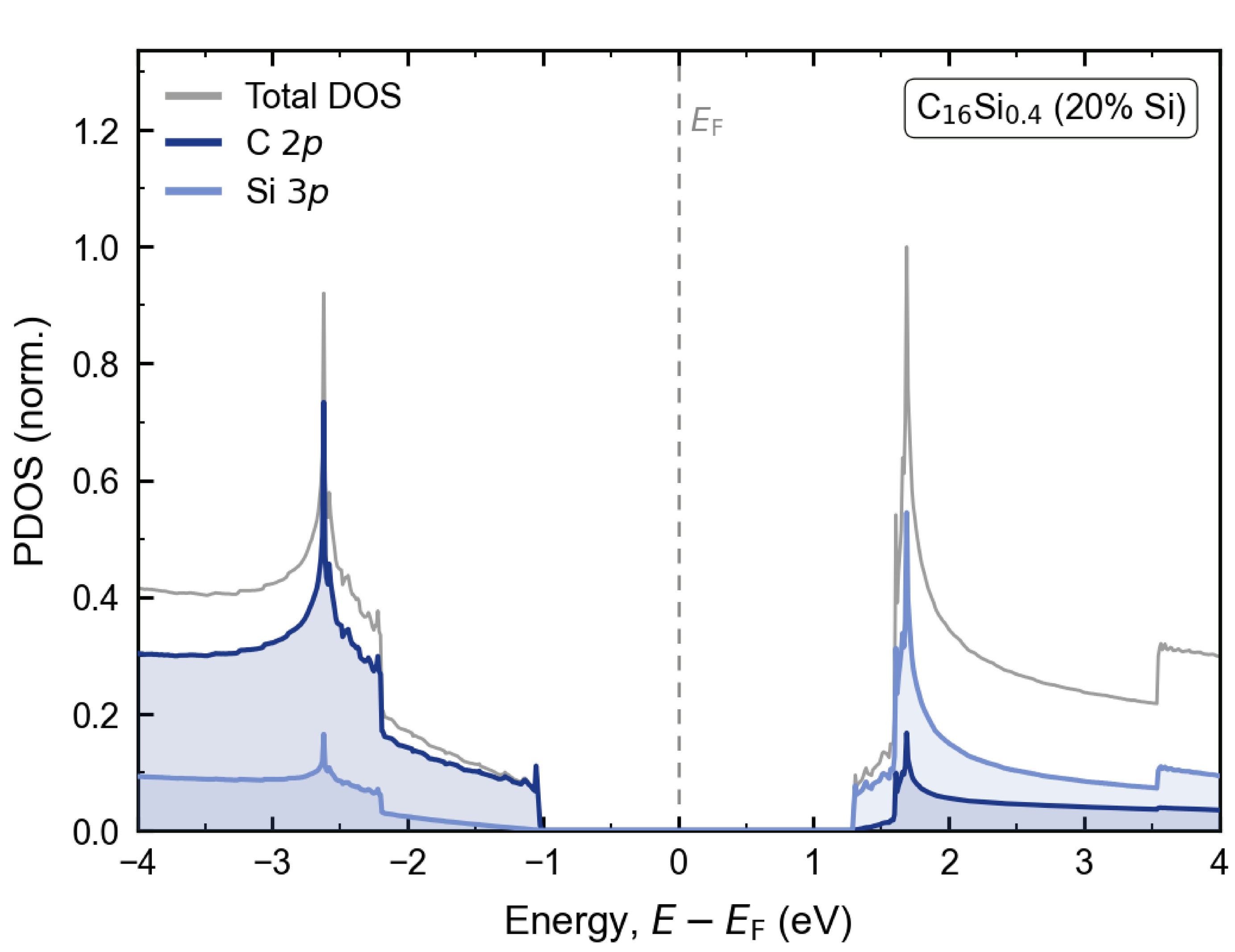
As conventional silicon-based device scaling approaches fundamental physical limits, two-dimensional carbon-based materials have emerged as promising candidates for next-gen nanoelectronic applications. Graphene's unique electronic structure allows for diverse properties that can be engineered with silicon-doping.

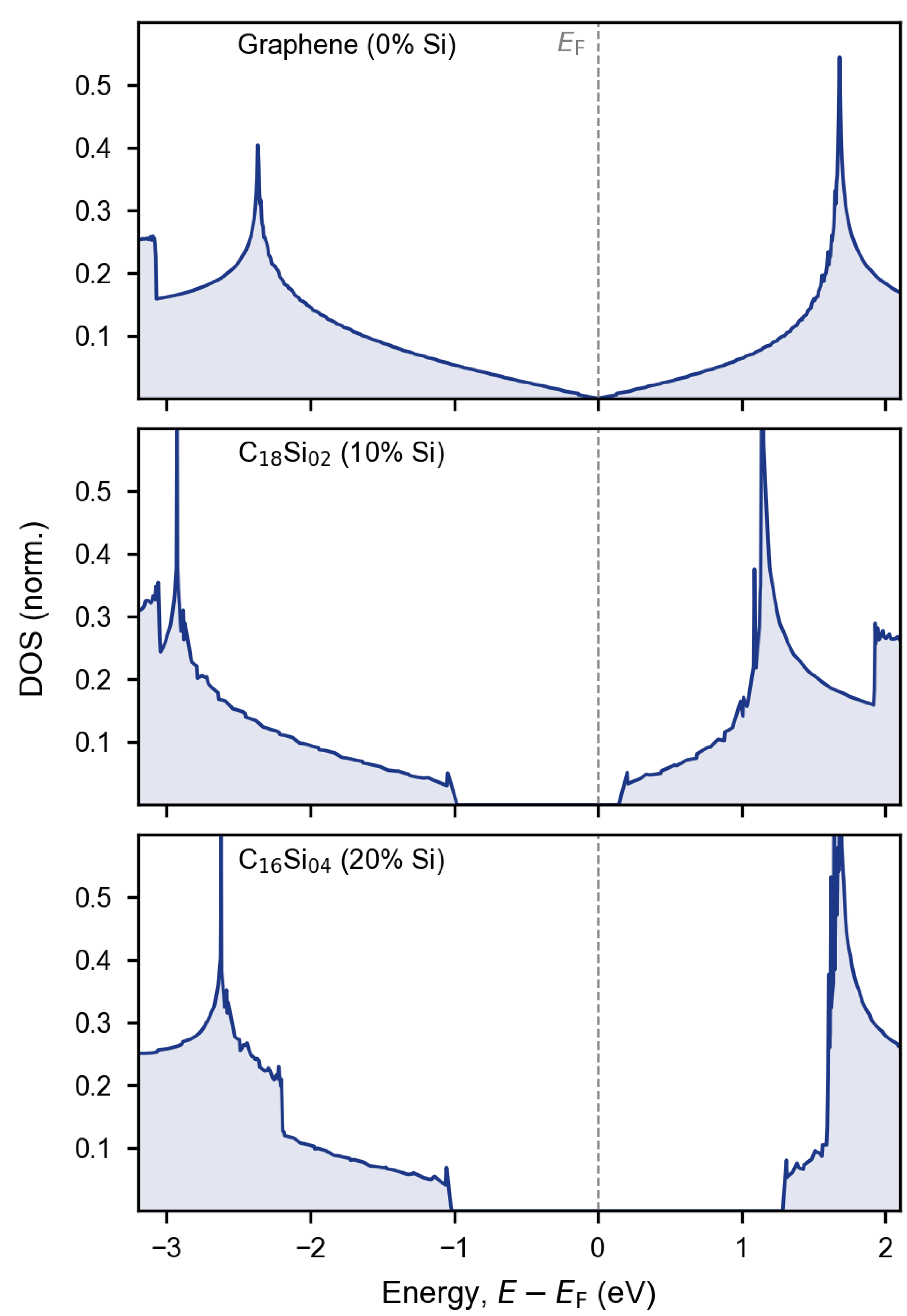
Materials and Methodology

Density Functional Theory (DFT) calculations were carried out using Quantum ESPRESSO with the PBE (GGA) approximation. Ultrasoft pseudopotentials and a plane-wave basis set were employed, with a kinetic energy cutoff of 40 Ry. Silicon doping was modeled using the Virtual Crystal Approximation (VCA) via virtual2.x module. A 60×60×1 k-point grid and the tetrahedron method were used for electronic structure calculations. The Density of States (DOS) and Projected Density of States (PDOS) were obtained using the dos.x and projwfc.x utilities, and structural relaxation was performed using vc-relax calculations.

Results







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Conclusions

These preliminary results highlight the potential of silicon doping as a tunable parameter for graphene-based nanoelectronic systems. The observed trend suggests progressive distortion of the Dirac cone and the emergence of bandgap opening with increasing silicon concentration, indicating a transition toward semiconducting behavior.